

Use the taught distinction without being walked through it first.

Open-ended consolidation question

A colourless gas is evolved. When a lighted splint is brought near the gas, it burns with a 'pop' sound.

Write a complete evidence-to-claim response. Your answer must clearly separate what was observed or tested from what you infer chemically.

Your response:

SELF-GUIDED HELP - look only after attempting

1. VISUAL / LOGIC CUE

> Separate 'what happened' from 'what it means'.

Observation/test belongs on the left.

Gas identity belongs on the right.

2. WORKING FRAME

Observation: gas evolved.

Test: lighted splint gives a pop sound.

Inference: the gas is hydrogen.

Evidence link: pop test supports H₂.

3. ANSWER + CHECK

Answer:

The gas is hydrogen.

The inference is supported by the pop test.

Check: the claim does not exceed the evidence.

Student response

"I observed hydrogen gas because the gas burned with a pop sound."

The chemistry conclusion may be right, but the sentence mixes observation and inference.

Open-ended repair

Rewrite the response so that observation/test, inference, and evidence link are separate. Then explain exactly what was wrong with the original wording.

SELF-GUIDED HELP - look only after attempting

1. VISUAL / LOGIC CUE

> Ask: could the first clause be seen directly?

'Hydrogen' is not directly observed.
The pop is the test evidence.

2. WORKING FRAME

Observation/test: gas burns with a pop sound.
Inference: gas is hydrogen.
Link: pop test supports the hydrogen inference.

3. ANSWER + CHECK

Answer:

Original error: inference was written as observation.

Check: evidence is stated before the chemical claim.